
REAL-TIME AI-BASED SURVEILLANCE SYSTEM FOR MOBILE PHONE DETECTION IN RESTRICTED AREAS

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B.Tech AIML

Engineering Design Project



PROBLEM STATEMENT

Problem

- Manual CCTV monitoring is inefficient
- Human errors can miss violations
- Mobile phone misuse goes unnoticed

Importance

- Need automated real-time monitoring
- Improves security and reduces manual effort



OBJECTIVES

- Detect persons and mobile phones in real time
- Identify unauthorized mobile usage
- Generate instant alerts
- Save screenshots as evidence
- Maintain event logs
- Deploy on Jetson Nano

METHODOLOGY / APPROACH

Block Diagram

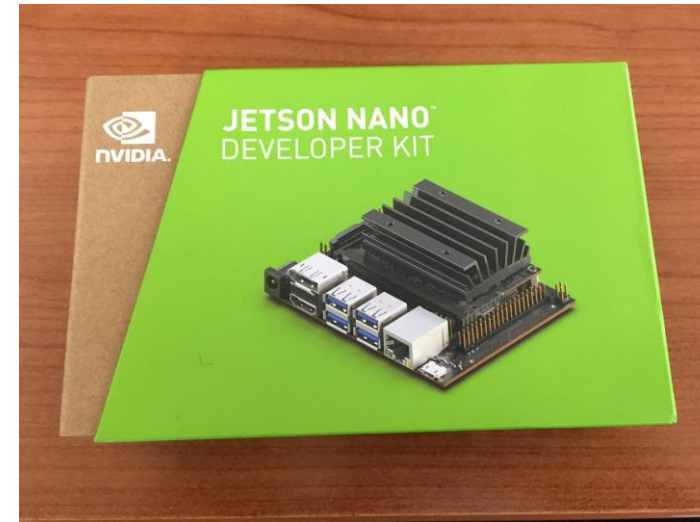


Workflow

Live video is captured, processed frame by frame, checked for person + mobile phone, then alert and evidence are generated.

Technologies Used

- Jetson Nano
- YOLO
- OpenCV
- Python



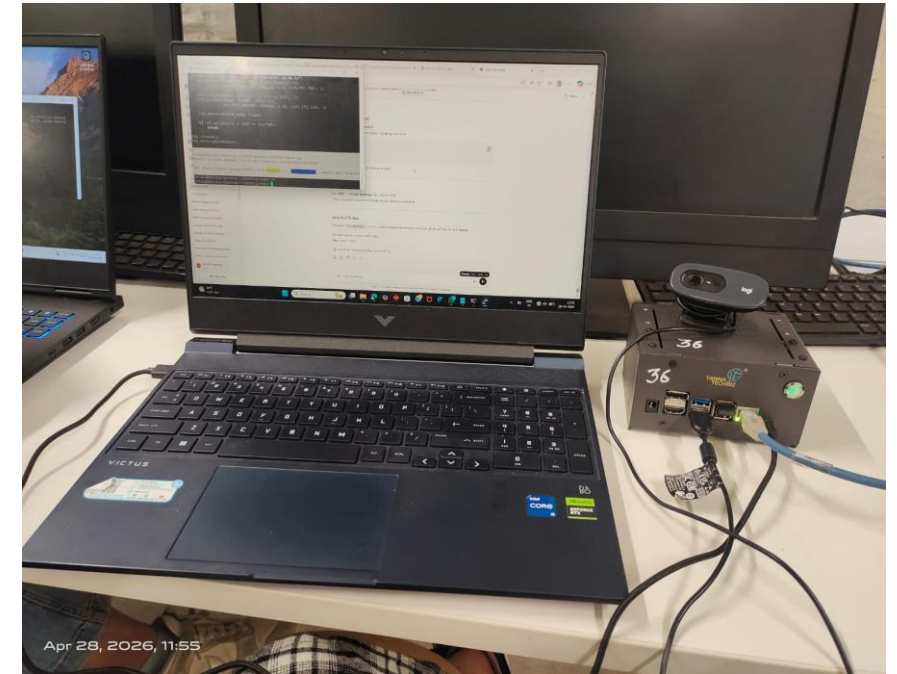
HARDWARE & SOFTWARE REQUIREMENTS

Hardware

- NVIDIA Jetson Nano
- USB Camera / Webcam
- Monitor
- Keyboard & Mouse

Software

Python
OpenCV
YOLO
VS Code



IMPLEMENTATION

- **STEPS FOLLOWED**

- Installed OpenCV, NumPy, and required libraries
- Loaded MobileNetSSD pre-trained model
- Captured live video using camera
- Applied AI model for object detection on each frame
- Displayed results with bounding boxes
- Saved detected events in CSV file
- **System Architecture**
- Camera → Frame Capture → AI Model (Detection) → Output Display + Logging

CODE

```
SQL (4).py:nb
import cv2
import os
import csv
import time
from datetime import datetime

SAVE_DIR = "violations"
LOG_FILE = "logs.csv"
CAM_INDEX = 0
WINDOW_NAME = "AI Surveillance System"
ALERT_COOLDOWN = 3

os.makedirs(SAVE_DIR, exist_ok=True)

if not os.path.exists(LOG_FILE):
    with open(LOG_FILE, "w", newline="") as f:
        writer = csv.writer(f)
        writer.writerow(["Timestamp", "Event", "Image"])

cap = cv2.VideoCapture(CAM_INDEX)
cap.set(3, 960)
cap.set(4, 540)

fgbg = cv2.createBackgroundSubtractorMOG2(history=500, varThreshold=50)
last_alert = 0

ZONE_X1, ZONE_Y1 = 220, 100
ZONE_X2, ZONE_Y2 = 740, 450

while True:
    ret, frame = cap.read()
    if not ret:
        print("Camera disconnected. Restarting...")
        cap.release()
        time.sleep(2)
        cap = cv2.VideoCapture(CAM_INDEX)
        continue
    7 12 Downloading update...
```

```

gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
fgmask = fgbg.apply(gray)

_, thresh = cv2.threshold(fgmask, 200, 255, cv2.THRESH_BINARY)
_, contours, _ = cv2.findContours(thresh, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

person_found = False

for cnt in contours:
    area = cv2.contourArea(cnt)
    if area < 5000:
        continue

    x, y, w, h = cv2.boundingRect(cnt)

    if x > ZONE_X1 and y > ZONE_Y1 and x+w < ZONE_X2 and y+h < ZONE_Y2:
        person_found = True
        cv2.rectangle(frame, (x, y), (x+w, y+h), (0, 255, 0), 2)
        cv2.putText(frame, "Person Detected", (x, y-8),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.55, (0, 255, 0), 2)

cv2.rectangle(frame, (ZONE_X1, ZONE_Y1), (ZONE_X2, ZONE_Y2), (255, 0, 0), 2)
cv2.putText(frame, "Restricted Zone", (ZONE_X1, ZONE_Y1 - 10),
            cv2.FONT_HERSHEY_SIMPLEX, 0.6, (255, 0, 0), 2)

if person_found:
    cv2.rectangle(frame, (0, 0), (frame.shape[1], 50), (0, 0, 255), -1)
    cv2.putText(frame, "ALERT: UNAUTHORIZED MOVEMENT DETECTED",
                (20, 32), cv2.FONT_HERSHEY_SIMPLEX, 0.75,
                (255, 255, 255), 2)

    if time.time() - last_alert > ALERT_COOLDOWN:
        ts = datetime.now().strftime("%Y-%m-%d_%H-%M-%S")
        img_name = os.path.join(SAVE_DIR, f"violation_{ts}.jpg")
        cv2.imwrite(img_name, frame)

```

Downloading update: 34 MB / 150.9 MB • 23%

Ln 99, 0


```

        cv2.imwrite(img_name, frame)

        with open(LOG_FILE, "a", newline="") as f:
            writer = csv.writer(f)
            writer.writerow([ts, "Restricted zone violation", img_name])

        last_alert = time.time()

    cv2.rectangle(frame, (0, frame.shape[0]-35), (frame.shape[1], frame.shape[0]), (30, 30, 30), -1)
    cv2.putText(frame, "AI Surveillance Monitor", (15, frame.shape[0]-10),
                cv2.FONT_HERSHEY_SIMPLEX, 0.6, (255,255,255), 2)

    now = datetime.now().strftime("%d-%m-%Y %H:%M:%S")
    cv2.putText(frame, now, (720, frame.shape[0]-10),
                cv2.FONT_HERSHEY_SIMPLEX, 0.5, (200,200,200), 1)

    cv2.circle(frame, (900, 20), 8, (0, 0, 255), -1)
    cv2.putText(frame, "LIVE", (915, 25),
                cv2.FONT_HERSHEY_SIMPLEX, 0.55, (255,255,255), 2)

    cv2.imshow(WINDOW_NAME, frame)

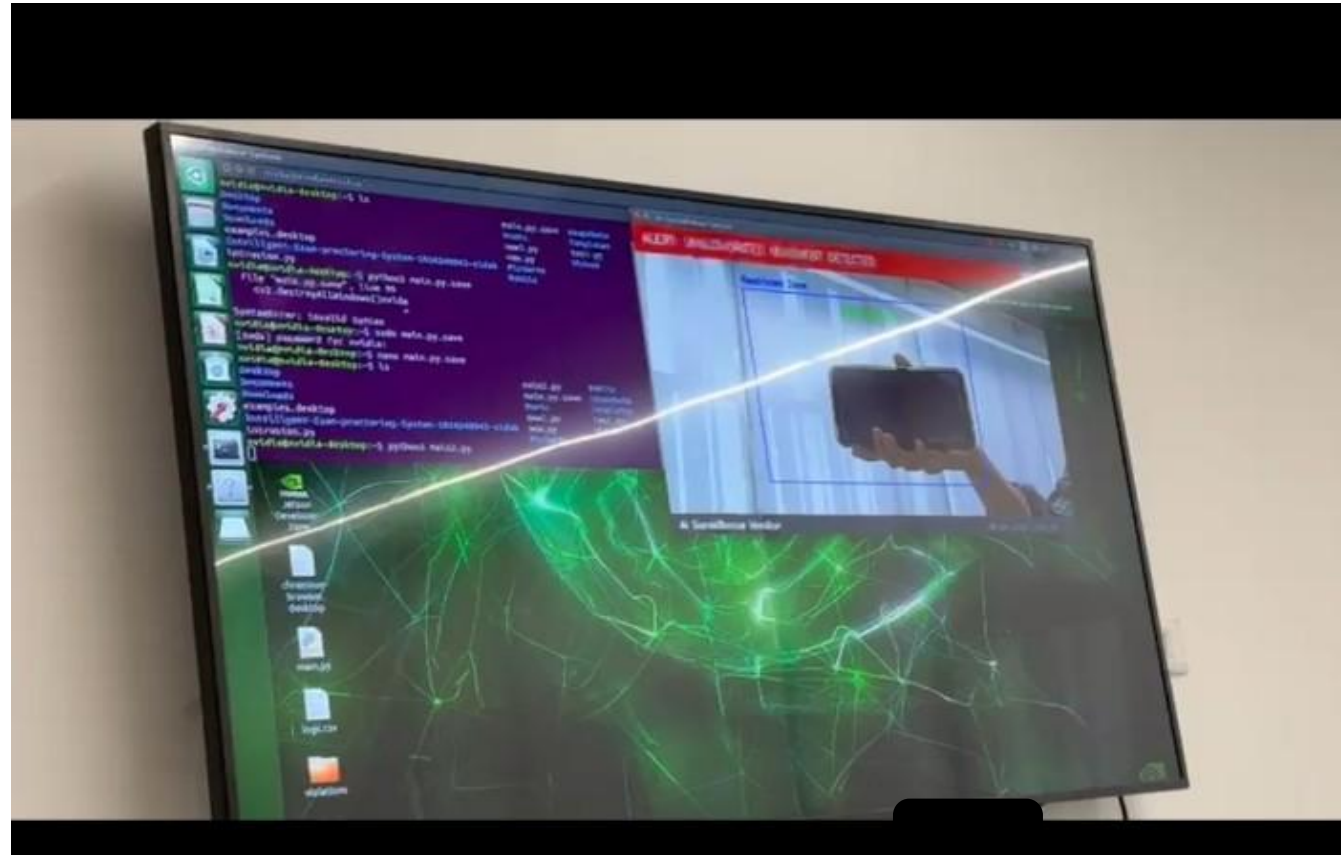
    if cv2.waitKey(1) & 0xFF == ord("q"):
        break

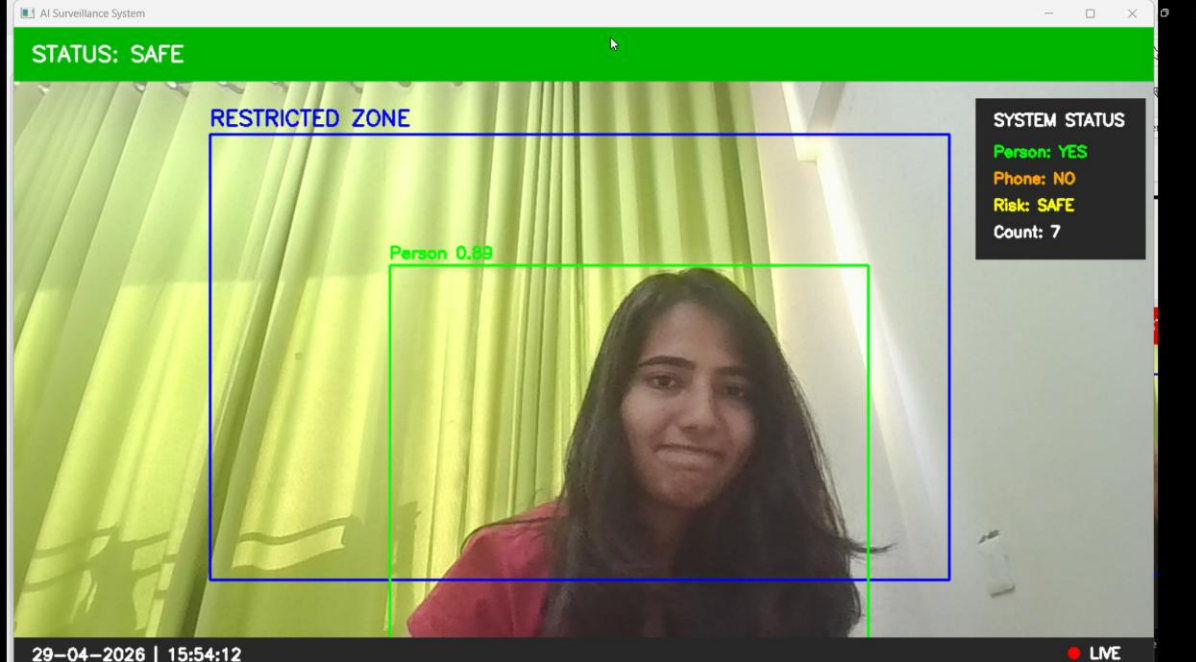
cap.release()
cv2.destroyAllWindows()

```

RESULTS:

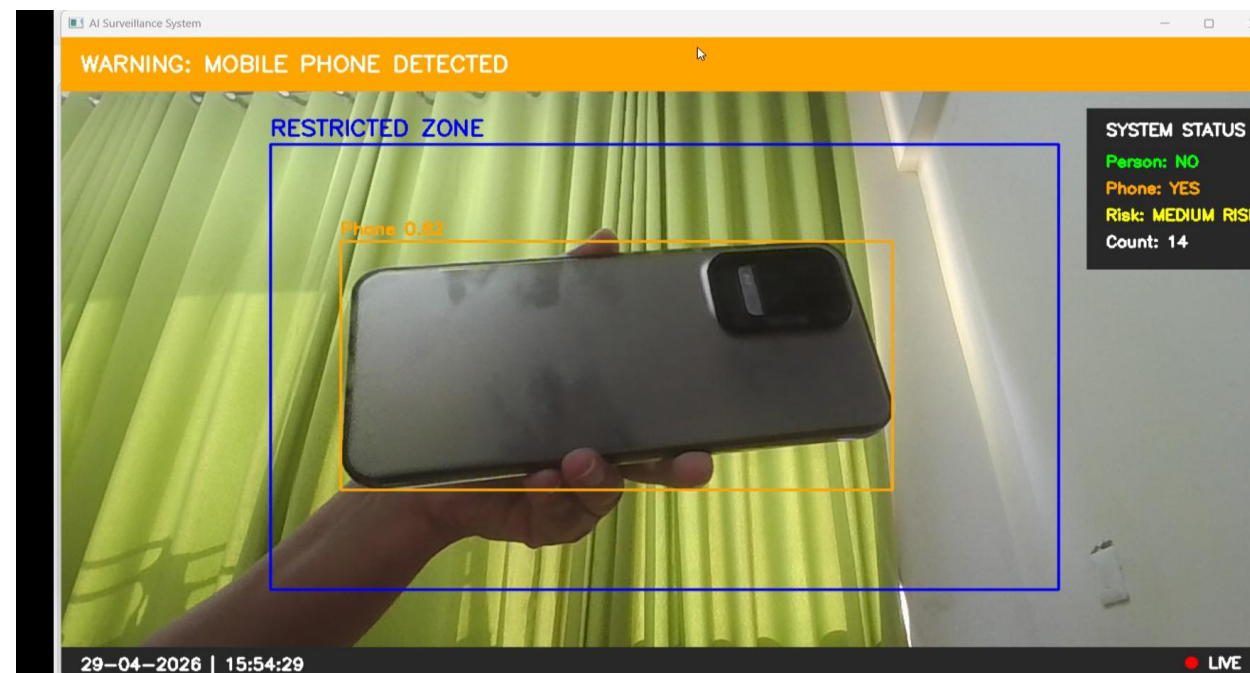
- On Jetson Nano





← Only Person detected
SAFE

Only Mobile Phone
detected
MEDIUM RISK



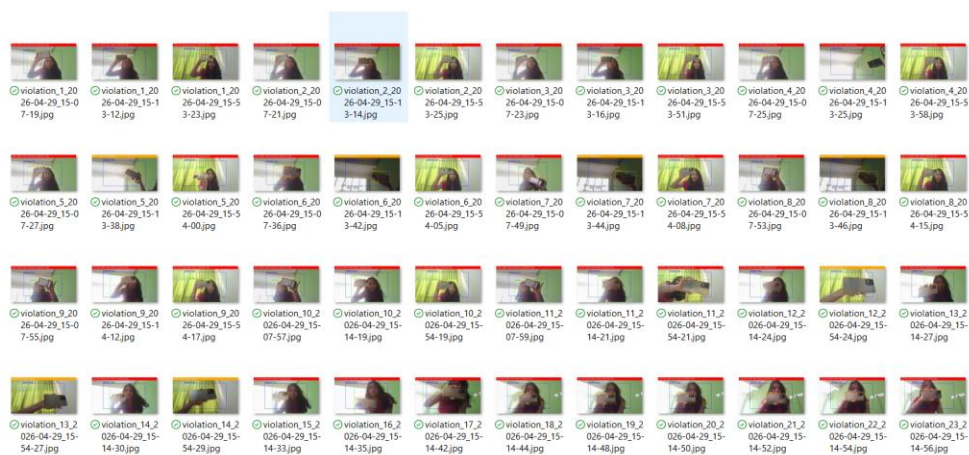
Person with Mobile Phone detected – HIGH RISK



The output is displayed in a CCTV-like interface with real-time detection, live status, and exact date-time monitoring

name	Status	Date modified	Type	Size
violations	✓	28-04-2026 22:42	File folder	
logs.csv	↻	28-04-2026 22:42	Microsoft Excel Co...	3 KB
main3.py	✓	28-04-2026 22:34	PY File	5 KB
yolov8n.pt	✓	28-04-2026 22:38	PT File	6,397 KB

Recorded screenshots



	A	B	C	D	E	F	G	H
1	Timestamp	Event	Image					
2	2026-04-2	Phone dete	violations\violation_1_2026-04-28_22-40-52.jpg					
3	2026-04-2	Phone dete	violations\violation_2_2026-04-28_22-40-54.jpg					
4	2026-04-2	Phone dete	violations\violation_3_2026-04-28_22-40-56.jpg					
5	2026-04-2	Phone dete	violations\violation_4_2026-04-28_22-41-04.jpg					
6	2026-04-2	Phone dete	violations\violation_5_2026-04-28_22-41-07.jpg					
7	2026-04-2	Phone dete	violations\violation_6_2026-04-28_22-41-09.jpg					
8	2026-04-2	Phone dete	violations\violation_7_2026-04-28_22-41-24.jpg					
9	2026-04-2	Phone dete	violations\violation_8_2026-04-28_22-41-26.jpg					
10	2026-04-2	Phone dete	violations\violation_9_2026-04-28_22-41-28.jpg					
11	2026-04-2	Phone dete	violations\violation_10_2026-04-28_22-41-58.jpg					
12	2026-04-2	Phone dete	violations\violation_11_2026-04-28_22-42-00.jpg					
13	2026-04-2	Phone dete	violations\violation_12_2026-04-28_22-42-02.jpg					
14	2026-04-2	Phone dete	violations\violation_13_2026-04-28_22-42-20.jpg					
15	2026-04-2	Phone dete	violations\violation_14_2026-04-28_22-42-33.jpg					
16	2026-04-2	Phone dete	violations\violation_15_2026-04-28_22-42-35.jpg					
17	2026-04-2	Phone dete	violations\violation_16_2026-04-28_22-42-37.jpg					
18	2026-04-2	Phone dete	violations\violation_17_2026-04-28_22-42-39.jpg					
19	2026-04-2	Phone dete	violations\violation_18_2026-04-28_22-42-47.jpg					
20	2026-04-2	Phone dete	violations\violation_19_2026-04-28_22-42-49.jpg					
21	2026-04-2	Phone dete	violations\violation_20_2026-04-28_22-42-51.jpg					
22	2026-04-2	Phone dete	violations\violation_21_2026-04-28_22-42-53.jpg					

Logs the violations in csv file

PERFORMANCE

- **Detection Accuracy:** ~75–85% in normal lighting
- **Processing Speed:** ~15–20 FPS on laptop webcam
- **Response Time:** Alert generated within 1–2 seconds
- **System Reliability:** Works well in standard indoor conditions

CHALLENGES & SOLUTIONS

CHALLENGES:

- Slow processing on Jetson Nano
- Camera initialization issues
- Model loading delay

• Solutions:

- Used lightweight YOLOv8n model
- Optimized frame processing
- Ensured correct camera configuration

APPLICATIONS

- Examination hall monitoring
- Laboratory surveillance
- Office security systems
- Industrial restricted zones
- Smart campus monitoring systems



CONCLUSION & FUTURE SCOPE

- Conclusion:
 - The project successfully demonstrates an AI-based surveillance system capable of detecting unauthorized mobile phone usage in restricted areas using real-time video processing on Jetson Nano.
- Future Scope:
 - Face recognition integration
 - Cloud-based monitoring system
 - Multi-camera surveillance
 - Mobile alert notifications
 - Behavior analysis and prediction



